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Can a sepsis warning built only from what clinicians knew at the time stay useful in a hospital it has never seen?

PhD: Development of Novel AI Models for Early Detection and Prediction of Infectious Diseases, with a Focus on Sepsis

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Sepsis kills about 11 million people a year, and the hours before antibiotics decide who survives

49 M
cases of sepsis each year worldwide

11 M
deaths each year, 1 in 5 of all deaths

+4 to 8 %
higher mortality for every hour antibiotics are delayed

Sepsis-3 definition

A suspected infection plus new organ dysfunction: a rise of 2 or more points on the SOFA organ-failure score. "Onset" is the hour both are met.

Infection starts
often unnoticed

Physiology drifts
heart rate, breathing, labs change hour by hour

Our warning target
6 hours before onset

Sepsis onset
organ dysfunction meets the definition

Recognition and antibiotics
the clock the mortality figures run on

The pharmacist's view: early antibiotics save lives, but antibiotics "just in case" for everyone breed resistance. A warning is only useful if it is right often enough to act on, and early enough to matter.

Rudd et al., Lancet 2020 (global burden); Singer et al., JAMA 2016 (Sepsis-3); Kumar et al., Crit Care Med 2006 and Seymour et al., NEJM 2017 (hourly delay and mortality).

Published sepsis AI scores well on its own data and fails where it matters: the next hospital

PhysioNet 2019 Challenge: 104 teams, one scoring rule

0.36

winning utility on the two hospitals in training

-0.12

same model on the hidden third hospital: worse than no alarm at all

Utility: the challenge score that rewards an alarm 6 h before onset and penalises false alarms (0 = never alarming).

Epic Sepsis Model: deployed in hundreds of US hospitals

0.63

AUROC in independent validation, against 0.76 to 0.83 claimed by the vendor

67 %

of sepsis patients missed, while alerting on 18 % of all admissions

A systematic review of 28 models reports AUROC 0.68 to 0.99; almost none were tested prospectively.

Three things the literature rarely controls

Future information leaks into training

hours after onset labelled positive; values used before they were recorded; treatments that define the label used as inputs

Scored by ranking, not by whether an alarm helps

AUROC on one split of one database; no cost for false alarms, no credit for warning time

Each study defines sepsis its own way

results cannot be compared, and a model tuned to one definition fails under another

Reyna et al., Crit Care Med 2020 (challenge results); Wong et al., JAMA Intern Med 2021 (Epic external validation); Fleuren et al., Intensive Care Med 2020 (review).

The question the literature has not asked

Using only the information a clinician could have seen at each hour, can a model warn of sepsis six hours ahead with positive clinical utility, and does that utility survive transfer to a hospital system it has never seen?

What has been asked

How high can the AUROC go on one prepared dataset, with one team's label definition and whatever inputs were available?

What this project asks

Would the alarm have helped this patient, at an acceptable false-alarm burden, and would it still help in the next hospital, under one shared definition of sepsis?

RQ1 Predictors

Which information known at hour t carries the signal: vital signs and laboratory values alone, or also treatments, devices, admission context and comorbidity that no public model uses?

RQ2 Architecture

Once causality is enforced, does structuring the model around organ systems add anything over simpler sequence models? A hypothesis to test, not an assumption.

RQ3 Transfer and trust

Does positive utility in one ICU database survive in another, and what would make the clinicians who receive the alarm act on it?

Why a clinician building the model changes what gets built

The record is written by clinicians' decisions

Every order, culture and charted value exists because someone at the bedside decided to act. A model that does not know why a value exists learns the clinician's suspicion, not the patient's physiology, and that is exactly what fails in the next hospital.

A pharmacist among engineers

I can answer the clinical questions myself: which inputs are safe, which leak the answer, which alarm a team would act on. The engineering questions are where I need this room.

Example 1: antibiotics

What is usually done

Most pipelines either feed every antibiotic order to the model, which then predicts the label from itself because the label is defined by antibiotics plus cultures, or drop antibiotics altogether.

What we do instead

A pharmacist reads the order: cefazolin at induction of anaesthesia, stopped within 24 hours, no culture sent, is surgical prophylaxis, not suspicion of sepsis. We keep prophylaxis as a predictor by a guideline rule and exclude empirical and culture-directed antibiotics.

Example 2: clinician-suspicion proxies

What is usually done

A stress-dose hydrocortisone order or a chest X-ray request often means the team already suspects sepsis. To an engineer these are two strong predictors.

What we do instead

To a clinician they are the answer leaking in through the back door. We keep them, label them as proxies, and report every result with and without them, so the reader can see how much of the signal is the patient and how much is the team.

Prophylaxis rule from the ASHP / IDSA / SIS / SHEA surgical prophylaxis guidelines (2013) and the Surviving Sepsis Campaign (2021). The proxy sensitivity analysis is prespecified in the running comparison.

Why this way: six choices that go against common practice

Common practice	This project	Why
Challenge-prepared tables: PhysioNet 2019, 40 variables, two hospitals	**Raw MIMIC-IV 3.1 (one US academic centre, about 94,000 ICU stays) and eICU-CRD 2.0 (about 200 US hospitals, about 201,000 stays)**	Different hospitals and charting systems make the transfer test real
Each study defines sepsis its own way	**One Sepsis-3 definition applied identically to both databases (YAIB / ricu, Seymour suspicion window)**	Results are comparable across databases and with the benchmark literature
Values carried forward; hours after onset labelled positive	**Strict causality: each value used only from the hour it was recorded, expires after 24 h, with "measured?" and "how old?" channels; hours after onset excluded**	No information from the future reaches the model
40 to 125 variables	**655 clinician-defined predictors: vitals, labs, blood gases, SOFA / qSOFA / NEWS2, ventilation, vasopressor dose, fluids, urine output, devices, drug exposures, imaging orders, admission context, comorbidities**	The whole clinical picture, defined by someone who has treated these patients
Antibiotics used as predictors	**Antibiotics and cultures that define the label are excluded; prophylactic antibiotics are separated by a guideline rule and kept**	Avoids predicting the label from itself
AUROC on one split	**Utility, sensitivity at 0.5 or fewer false alarms per patient-day, warning time, calibration; 20 % of patients sealed and opened once**	Measures what would help patients, and cannot be tuned to the test

Key literature shaping these choices: Reyna 2020 (utility score); Seymour 2016 (suspicion window); van de Water 2024 (YAIB benchmark); Moor et al., eClinicalMedicine 2023 (training across sites improves transfer).

A mixed-methods design: the model, the clinicians who would use it, then hospital patients

Phase 1 | Quantitative (now)

Causal modelling on MIMIC-IV and eICU: one label definition, 655 predictors, transfer between databases, utility and false-alarm burden, a sealed holdout opened once.

Output: a warning with measured utility and its alarm policy.

Phase 2 | Qualitative (2027)

Interviews and a survey with ICU doctors, nurses and pharmacists: would they act on this alarm, what explanation do they need (overall risk or risk per organ), what false-alarm rate would they tolerate, what would make them trust or ignore it.

Output: the alarm policy and the explanation the model must meet.

Phase 3 | Future direction

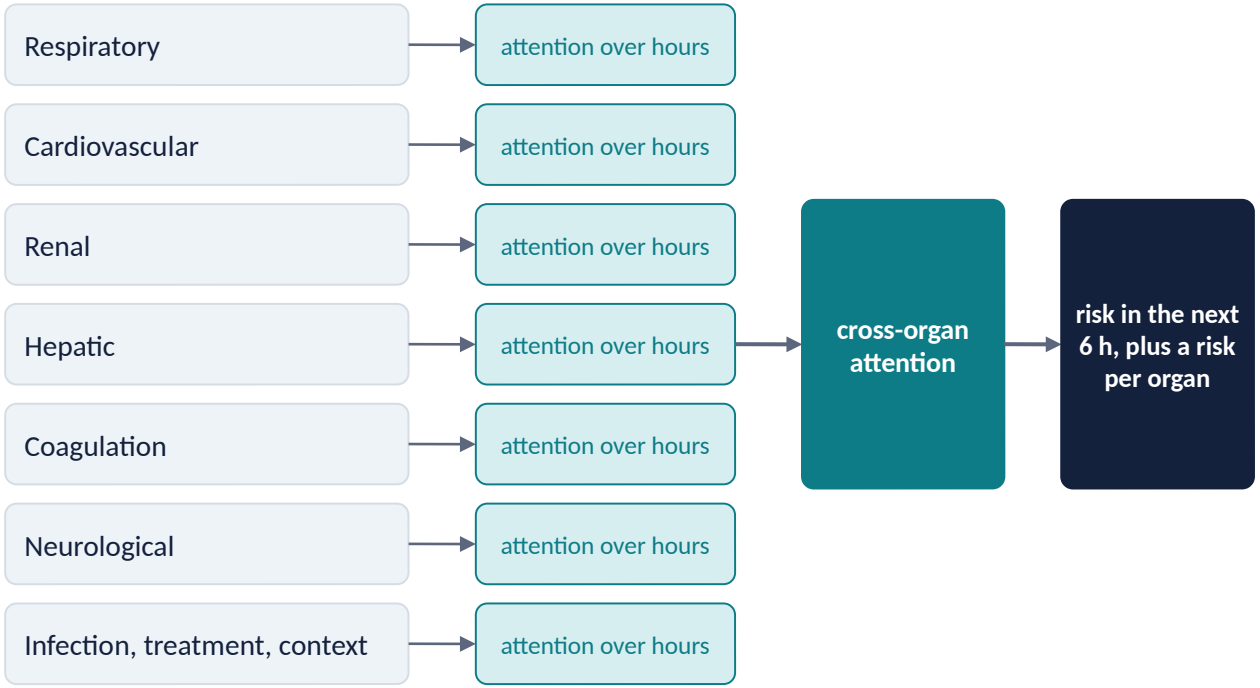
Test on hospital patients: the model runs silently on live ICU data, alarms are recorded but not shown, and compared with what happened. Then a clinician-facing pilot, with ethics approval and a data-governance agreement.

Output: evidence from real patients, in real time, in a hospital the model never trained on.

Why mixed: an alarm nobody trusts saves nobody. The clinicians' answers set the false-alarm budget and the explanation that the quantitative model has to meet before it reaches a patient. The pharmacist on the team can run that conversation in the ward's own language.

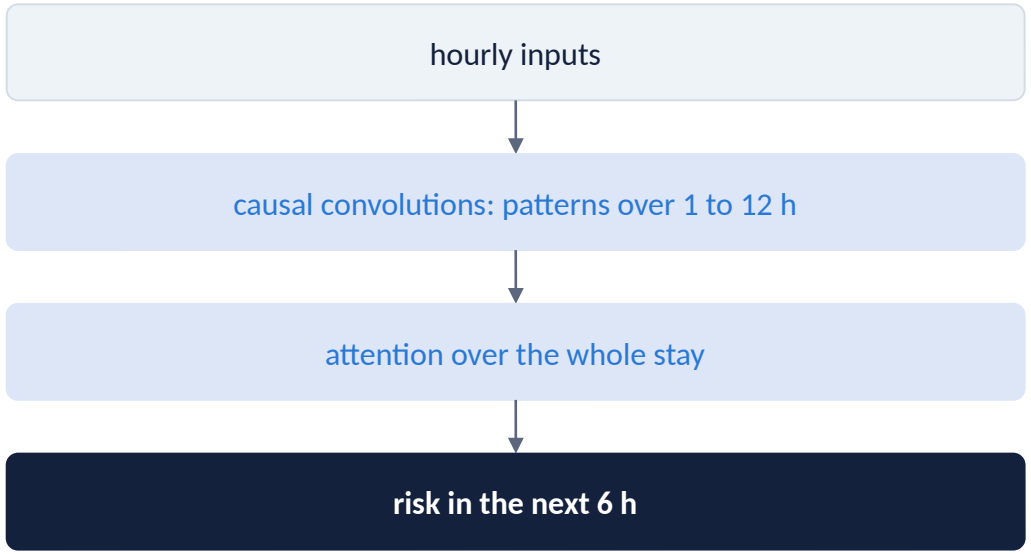
Organ-structured attention is a hypothesis we test, not an assumption

OSTA: organ-system-aware temporal attention (my architecture)



Each organ group is read by its own attention module, then the groups exchange information, mirroring how an ICU team reasons. Output: one overall risk and a risk per organ, so a warning can say which system is failing.

CNN-Transformer (my earlier architecture)



Compared on identical hours, decision rules fixed in advance

OSTA with and without organ grouping (the ablation that decides), GRU-D, temporal CNN, logistic regression, boosted trees, the challenge winner's method (path signatures + boosted trees), and self-supervised pretraining on 150,000 unlabelled stays. Organ structure is kept only if it beats the flat model on the validation split.

Preliminary result 1: removing information from the future is the biggest single correction

What an audit of our own pipeline found

- Hours after sepsis onset were labelled positive, so the model learnt to recognise sepsis that had already started.
- Measurements were carried forward from the time they were taken, not from the time they were available.
- Treatments that are part of the label (antibiotics, cultures) were present among the inputs.
- Earlier versions of our models were trained this way; they were retired and are not reported.

Numbers in the 0.9s in the literature should be read with this in mind.

Rebuilt causally: OSTA on the PhysioNet 2019 test set (3-seed ensemble)

0.81
patient-level AUROC

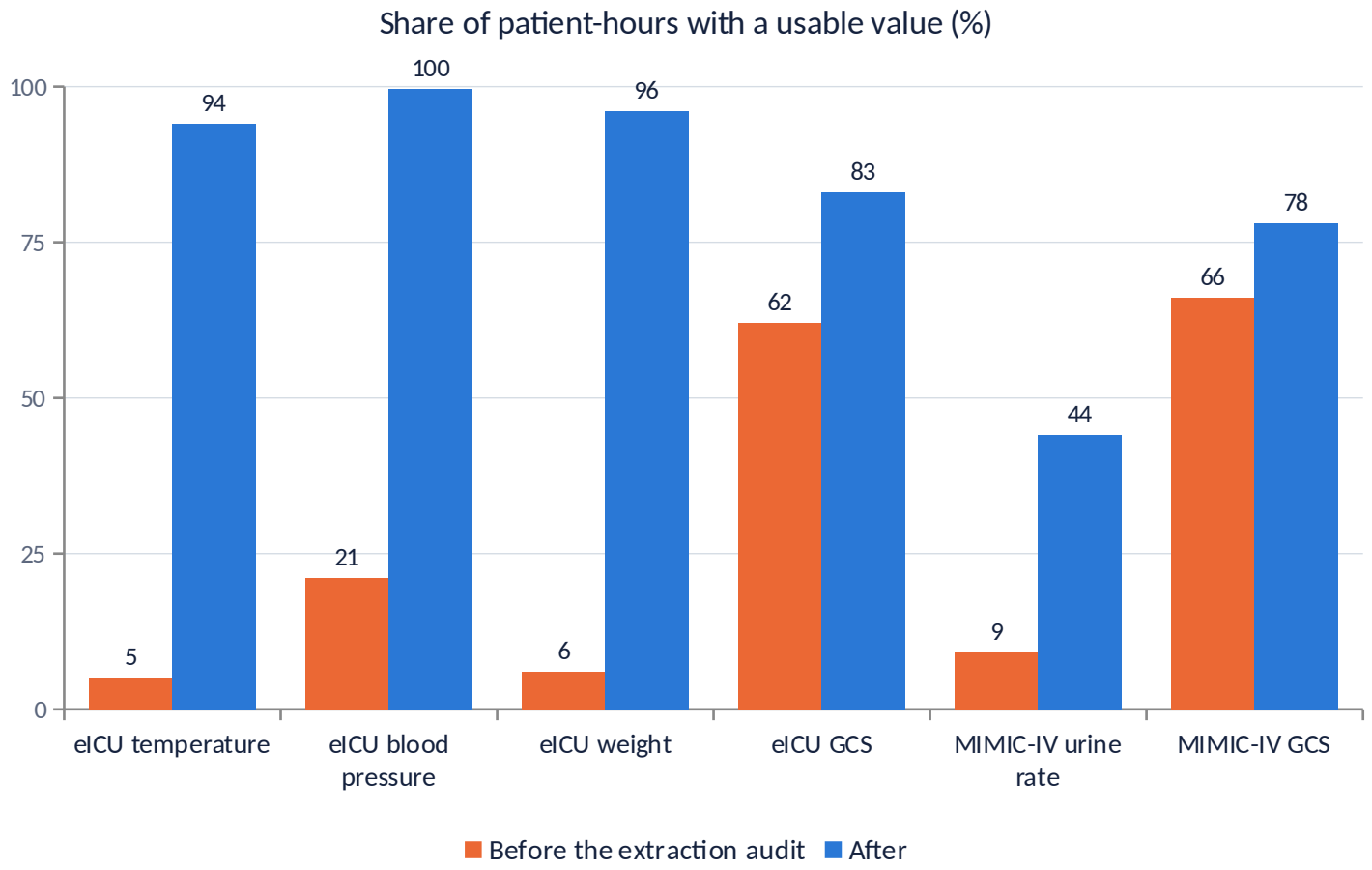
0.80
hourly AUROC

0.22
normalised utility (0.30 at the conformal threshold)

How to read these numbers

- Positive hours are 1.6 % of all hours: ranking scores look modest even for a useful model.
- Utility 0.22 means the alarms are worth about a fifth of a perfect alarm system, after paying for false alarms; at the conformal threshold (sensitivity 0.70, specificity 0.72) it is 0.30. The challenge winner scored 0.36 in-sample and below zero out of sample.
- This is the honest starting point from which every later number is measured.

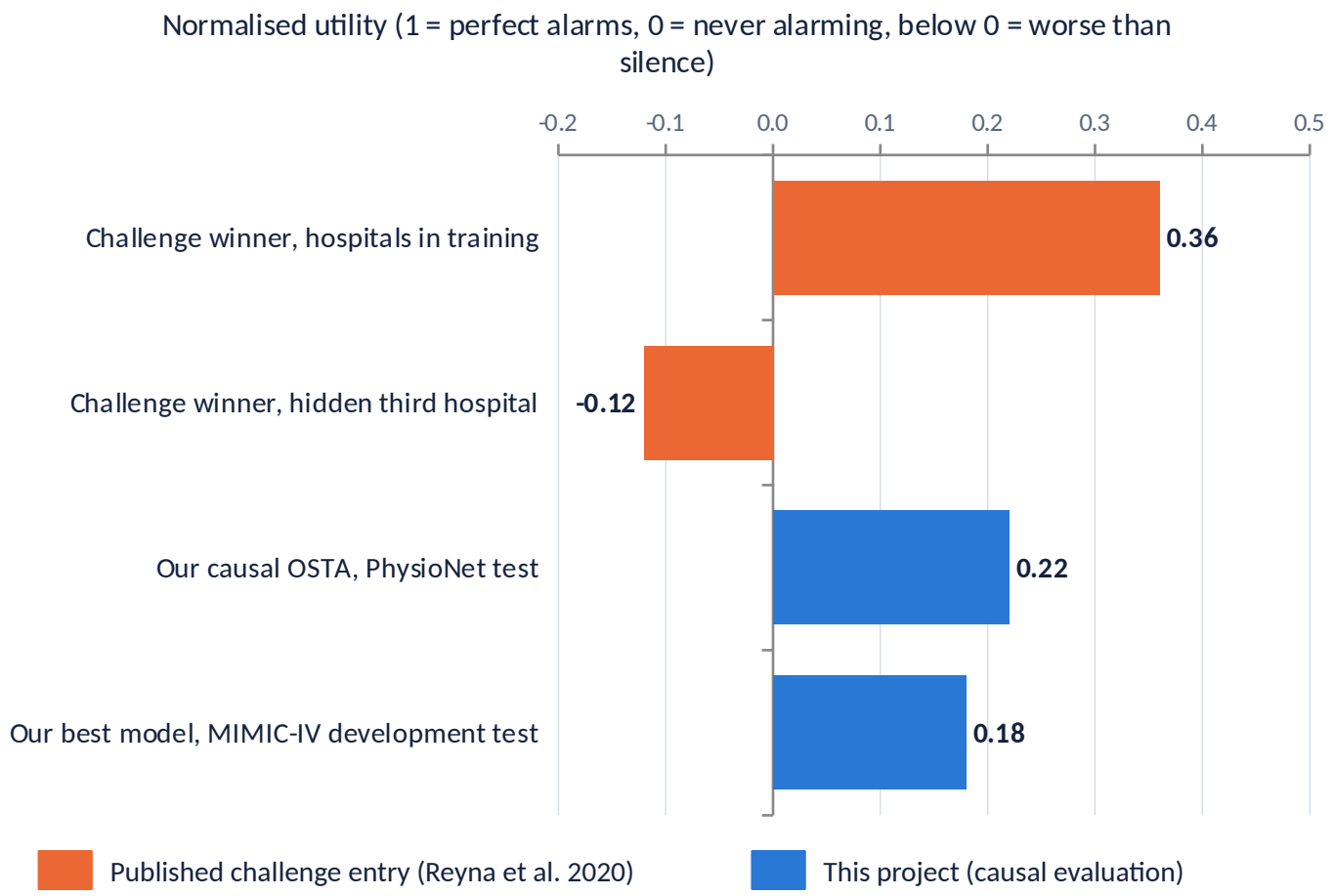
Preliminary result 2: making two raw databases speak the same language was most of the work



What the audit changed

- Samples of 10,000 ICU stays per database: 8,795 MIMIC-IV and 5,710 eICU stays after the benchmark's exclusions.
- 76 raw measurement channels mapped item by item, with every mapping checked against row counts and a unit audit.
- eICU nurse-charted vitals, cuff pressures and weights were in different tables from the ones most pipelines read: knowing how nurses chart found them.
- Urine output became a rate per kg only once each record was tied to the interval it covered, as the kidney criteria require.
- Nothing here is a model change; it is where the transfer problem actually lives.

Preliminary result 3: on MIMIC-IV the warning has positive utility, where the challenge winner collapsed on an unseen hospital



MIMIC-IV development test split

1,422 stays, 63,488 hours, 71 sepsis events; 0.65 % of hours are positive

0.81
hourly AUROC, 3 seeds (SD 0.01)

5 h
median warning before onset

1 in 2
sepsis events caught at 0.5 false alarms per patient-day

0.18
utility, 95 % CI 0.08 to 0.27

Development split of a 10,000-stay sample, 1 to 3 seeds. The sealed 20 % holdout has not been opened. Our run of the winner's own method on MIMIC-IV is in progress.

Preliminary result 4: on 10,000 stays every architecture overfits within three epochs, so the next lever is scale

What the training records show

1 to 3 epochs to the best validation score for every rich-feature neural model, while training loss keeps falling

175 septic stays in a training split: too few for 300 to 600 predictors

0.80 to 0.81 AUROC for OSTA, CNN-Transformer, output fusion and gated variants: differences smaller than seed noise (SD 0.012)

= 0 difference between OSTA with and without attention gates: the checkpoint is the first epoch either way

What this changes



Full cohort

about 94,000 MIMIC-IV and 201,000 eICU stays: roughly nine times more sepsis cases than the samples used so far, three seeds



Self-supervised pretraining

the encoder first learns from up to 150,000 unlabelled stays to forecast the next value and organ worsening, without sepsis labels



Loss chosen by utility

utility-weighted training against plain and time-smoothed losses, selected on the validation split by a rule written in advance



Running now

sixteen architectures, the organ-structure ablation and the challenge winner's method, on identical hours

The contribution so far is an honest, transferable test of sepsis warnings, built by a clinician

A causal, cross-database pipeline

one Sepsis-3 definition on two raw ICU databases; every value used from the hour it was known; 655 clinician-defined predictors; antibiotics handled by a guideline rule; proxies declared

Evaluation by what helps patients

utility, false alarms per patient-day and warning time with patient-level confidence intervals; a 20 % holdout sealed until the protocol is frozen; nothing tuned on test data

Three findings

leakage inflates published scores; data coverage drives transfer more than architecture; models overfit small samples within epochs

Three manuscripts submitted in 2026

AI in antimicrobial-resistance diagnostics (Int J Antimicrob Agents, Oct); temporal patterns and predictors of sepsis onset in 27,101 ICU admissions (Frontiers in Medicine, Sep); infrared thermography in infectious disease, systematic review with meta-analysis (JMIR, Jul)

The main study, on MIMIC-IV and eICU, is being written up for submission; the numbers shown today are its preliminary, development-split results.

Next steps: scale, the sealed holdout, the clinicians, and then hospital patients

Oct to Nov 2026	Sixteen-model comparison on identical hours; shortlist by the prespecified validation rule; full results into the manuscript
Nov to Dec 2026	Full cohort (about 94,000 + 201,000 stays), three seeds; leave-hospital-out evaluation across eICU's hospitals; pooled multi-site training
Jan 2027	Freeze the protocol and open the sealed holdout once; fix the alarm policy (sensitivity against false alarms per patient-day)
Feb to Apr 2027	Qualitative study: interviews and a survey with ICU doctors, nurses and pharmacists on trust, explanation and the alarm burden they would accept
2027 onward	Radiology reports as predictors (MIMIC-IV notes); fairness by subgroup; then the future direction: a silent prospective test on hospital patients, with ethics approval

Future direction: test the model on hospital patients

- The model runs on live ICU data; alarms are logged, not shown, and compared with what happened to each patient.
- Measures the three things no retrospective study can: real-time data delays, a hospital the model never trained on, and the alarm burden on a working unit.
- Then a clinician-facing pilot shaped by the qualitative study: who receives the alarm, what it says, what they do next.
- Needs ethics approval, a data-governance agreement and a clinical partner; the pharmacist on the team talks to the ward in its own language.

Where I would like your input



Which score should pick the model when they disagree?

Utility rewards an alarm six hours before onset and charges for every false-alarm hour; AUROC rewards ranking. On our data they already rank models differently. How much sensitivity would you trade to halve false alarms per patient-day, and should that trade-off be fixed before or after seeing the holdout?



How should I study trust before the model reaches a ward?

The qualitative phase asks ICU doctors, nurses and pharmacists whether they would act on the alarm and what they need to see. Interviews with patient vignettes, a survey with the model's own alarms, or a simulation with the alarm live on a screen: which would you choose, and what should it measure?

Thank you. Backup slides follow: glossary, the utility score, predictor catalogue, cohorts, the running comparison, and the RD6 supplement.

Glossary: the clinical terms used in this talk

Sepsis-3
Life-threatening organ dysfunction caused by a dysregulated response to infection (Singer et al. 2016). Operationalised as suspected infection plus a SOFA rise of 2 or more points.

Suspected infection
An antibiotic order and a culture within a 72 h / 24 h window of each other (Seymour et al. 2016). eICU has antibiotics only.

qSOFA and NEWS2
Bedside screening scores built from vital signs and mental state; used here as predictors, never as the label.

False alarms per patient-day
How often the alarm fires on patients who do not develop sepsis, expressed per 24 h of monitoring; 0.5 is the budget used in this project.

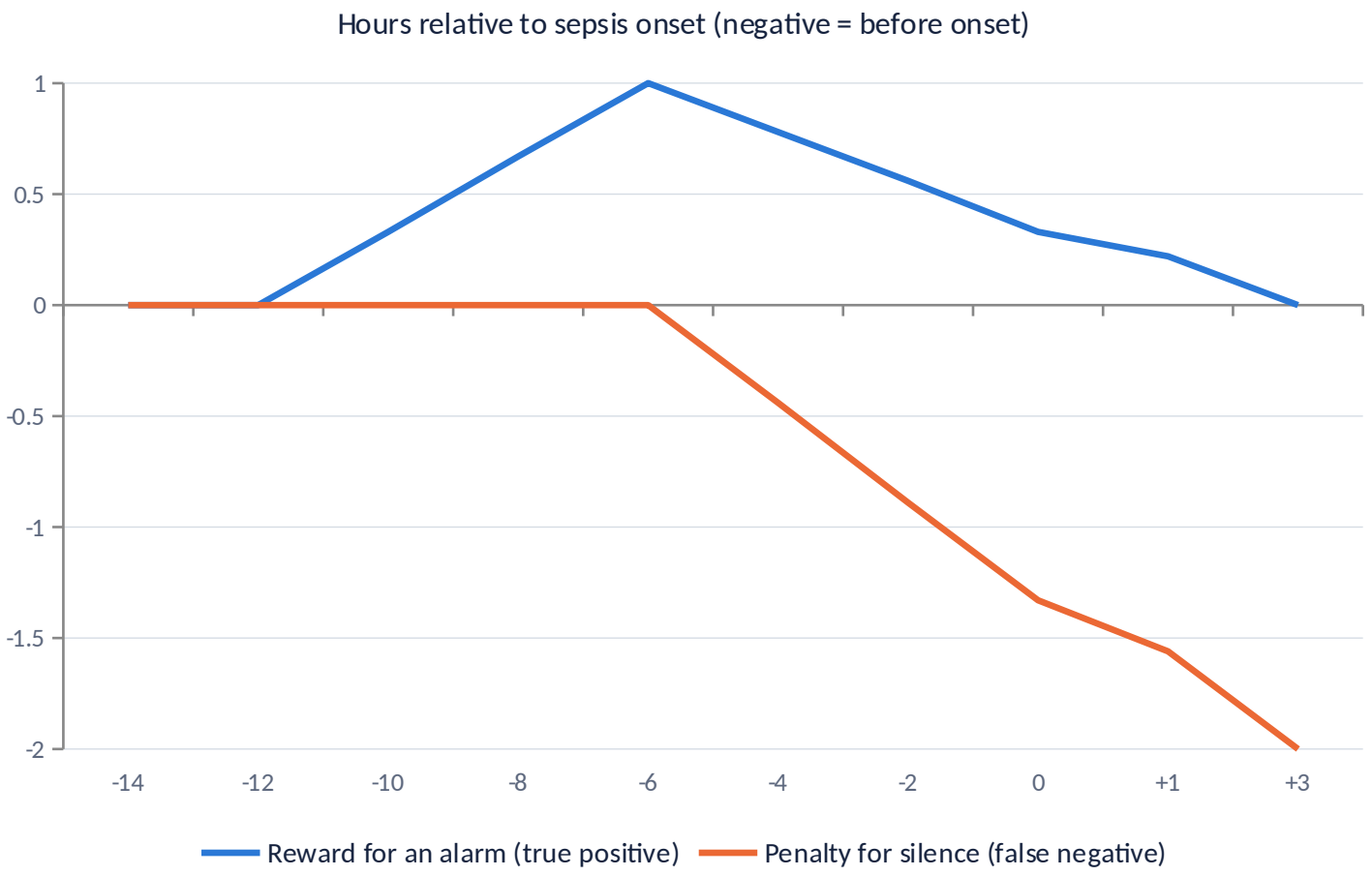
SOFA
Sequential Organ Failure Assessment: 0 to 4 points for each of six organ systems (respiratory, cardiovascular, renal, hepatic, coagulation, neurological).

Onset and horizon
Onset is the first hour both criteria are met. The model is asked, every hour, whether onset falls within the next 6 hours.

Utility score
PhysioNet 2019 scoring rule: reward for alarms from 12 h before to 3 h after onset, penalty for false-alarm hours and for missed sepsis, normalised so that 1 is perfect and 0 is never alarming.

Causal (availability-aware)
Each value is used only from the hour it was recorded in the system, with a measured-or-not flag and its age; no hour after onset is used for training or scoring.

The utility score: what an alarm is worth, hour by hour



Reading the score

- An alarm earns most when raised 6 h before onset; alarms earlier than 12 h earn nothing.
- Every false-alarm hour on a patient who never develops sepsis costs 0.05.
- Silence after onset is penalised up to 2 points per hour by 3 h after onset.
- Normalised utility = (score - score of never alarming) / (perfect score - score of never alarming).
- It rewards timing and punishes alert fatigue: the two things a bedside system is judged on.
- Official scorer code used unchanged; alarms evaluated hourly on the first 120 h of each stay.

The 655 predictors: the whole clinical picture, each with a measured flag and an age

Vital signs and monitoring

heart rate, blood pressures (cuff and arterial), respiratory rate, oxygen saturation, temperature, end-tidal CO2, central venous pressure, cardiac output, pain and sedation scores

Laboratory and blood gas

full chemistry, haematology with differential, coagulation, lactate, procalcitonin, troponin, BNP, ferritin, D-dimer, amylase, lipase, ammonia, urinalysis, blood gases

Scores and derived

SOFA (full and per organ), qSOFA, NEWS2, shock index, P/F and S/F ratios, neutrophil-lymphocyte ratio, BMI, changes against the previous value and the baseline, 24 h extremes and trends

Organ support

vasopressor dose in noradrenaline equivalents, ventilation mode and settings, oxygen flow, airway status, renal replacement therapy, urine output rate and AKI stage

Fluids, drugs and devices

crystalloids and transfusions, nutrition route, central and arterial lines, urinary catheters, drains; corticosteroids, immunosuppressants, chemotherapy, sedatives, opioids, diuretics, antipyretics

Context and history

admission source and ICU type, hours in hospital before ICU, prior admissions, prior positive cultures and antibiotics in earlier admissions, 17 comorbidity flags, infection-related phenotypes, imaging orders

Prophylactic antibiotics (MIMIC-IV)

surgical prophylaxis by agent, perioperative timing, duration and the absence of cultures or escalation; medical prophylaxis by agent and dose pattern

Never used as predictors

antibiotics and cultures that define suspected infection; race, ethnicity and insurance (fairness analysis only); anything recorded after the prediction hour

Cohorts, splits and the sealed holdout

Database	Candidate ICU stays	Sample used so far	After benchmark exclusions	Sealed holdout	Development use
MIMIC-IV 3.1 (BIDMC, Boston)	about 94,000	10,000	8,795 stays	20 % of patients by deterministic hash	train / validation / calibration / test, grouped by patient
eICU-CRD 2.0 (about 200 US hospitals)	about 201,000	10,000	5,710 stays	20 % of patients by deterministic hash	same, with hospital identifiers kept for leave-hospital-out

- Labels: Sepsis-3 via the YAIB benchmark (ricu), onset within the next 6 h; hours after onset excluded; stays followed for up to 120 h.
- Exclusions follow the benchmark (age, stay length, missing core data); the 10,000-stay samples are fixed random draws, identical across all runs so far.
- The holdout is released only once, after the full-cohort protocol is frozen; no threshold, calibration or model choice ever touches it.
- Transfer rows: each database's model is also scored on the other database's development patients.
- Label harmonisation is an open question: MIMIC-IV suspicion needs an antibiotic and a culture, eICU records antibiotics only; both definitions will be reported.
- Unlabelled pretraining corpus: up to 75,000 stays per database that are not in the labelled cohort and not in the holdout.

The comparison running now: sixteen models, three losses, rules fixed in advance

Group	Models	Purpose
Kept from earlier runs	OSTA, output fusion, CNN-Transformer, logistic regression, boosted trees	continuity with the 10,000-stay results
Literature benchmark	path signatures + boosted trees (PhysioNet 2019 winner's method)	direct replication of the winner on our data
New supervised	flat OSTA (same model, no organ grouping), GRU-D, causal temporal CNN, CNN encoder + OSTA head, event attention + OSTA head	organ-structure ablation; irregular-sampling models from the literature
Self-supervised	CNN encoder pretrained on unlabelled stays (next value, measured-within-6 h, SOFA components and organ worsening at 6/12/24 h), fine-tuned with OSTA, flat and survival (hazard) heads; site and pooled encoders	label-free learning from 150,000 stays; survival framing of the horizon
Losses	utility-weighted cross-entropy (primary), plain cross-entropy, temporal label smoothing	chosen by validation utility + AUPRC rank-average
Decision rules	organ structure kept only if OSTA beats flat OSTA on validation; shortlist = top 2 to 3 new models + baselines; full cohort with 3 seeds	no choice is made on test data

Evaluation additions in this run: warning time relative to the first antibiotic order, transfer to the other database, a conformal alarm policy with abstention on out-of-distribution hours, and a sensitivity analysis without clinician-suspicion proxies (hydrocortisone, imaging orders, prophylaxis).

RD6 supplement: publications and outputs, October 2025 to October 2026

Manuscript	Journal	Submitted	Status
Artificial Intelligence in the Diagnosis of Antimicrobial Resistance: Rapid Diagnostics and Imaging Innovations (narrative review)	International Journal of Antimicrobial Agents	9 Oct 2026	Technical check
Temporal Patterns and Predictors of Sepsis Onset in ICU Patients: A Retrospective Cohort Study Using PhysioNet Data	Frontiers in Medicine	23 Sep 2026	Editorial assignment
Diagnostic Applications of Infrared Thermography Across Infectious Diseases: A Systematic Review with a Subgroup Meta-Analysis	JMIR	18 Jul 2026	In review
Early sepsis warning on MIMIC-IV and eICU: causal predictors, utility and transfer (main study)	target: critical care informatics journal	in preparation	full-cohort run and sealed-holdout evaluation pending

- The cohort study (27,101 ICU admissions, 8 % sepsis, median onset 100 h) is the statistical companion to the modelling work: Cox models with a 6 h backshifted label identify early predictors that the deep models then use.
- The CNN-Transformer and OSTA manuscripts drafted in early 2026 were withheld after the leakage audit; both architectures are being re-evaluated causally in the current comparison.

RD6 supplement: training completed and what became of the March 2026 plan

Training and professional development

Harvard PPCR: Clinical Trials Design, Implementation and Analysis

completed, March 2025 to February 2026; informs the validation and alarm-policy study design

Iowa State University: Machine Learning Programme

completed, October to December 2025

DataCamp Python Data Science Career Track

in progress since September 2025

Research data and compute

PhysioNet credentialed access to MIMIC-IV 3.1 and eICU-CRD 2.0; reproducible notebook pipeline with 212 automated engineering checks

What happened to the March 2026 plan

MIMIC-IV external validation	became the core of the programme, extended to eICU with one shared label definition
Submit the CNN-Transformer and OSTA papers	paused after the leakage audit; both models are in the causal comparison instead
AMR review	revised and submitted to the International Journal of Antimicrobial Agents
Present at a conference	this conference; the RD6 review follows
Clinical stakeholder study	now the qualitative phase of a mixed-methods design, scheduled after the full-cohort results